```
  org 000bh                ;timer0 inter-
rupt vector address
      clr tr0              ;clear timer0
run bit
      mov tl0,#00h
      mov th0,#00h         ;reload timer0
with initial count
      djnz r7,a1
      mov r7,#46
      cjne r1,#60h,a5  ;check to again
start entering from left-side
      sjmp a4
  a5: cjne r1,#50h,a2 ;check for display
to stay on reaching display-16
      sjmp a3
  a2: inc r1
      sjmp a1
  a3: djnz r6,a1
      inc r1
      sjmp a1
  a4: mov r6,#10
      mov r1,#41h
  a1: setb tr0             ;set timer0 run
bit
      reti                 ;return from
timer0 ISR and clear tf0

  main: mov ie,#00h
        setb ea            ;set global
interrupt bit
        setb et0           ;enable
timer0 interrupt
      mov tmod,#01h        ;timer0 con-
figured in mode 1
      mov tcon,#00h
      mov tl0,#00h
      mov th0,#00h         ;set initial
count to 0000H
      mov r7,#46           ;provides gap
between each shift
      mov r6,#10     ;

      mov r0,#60h
  blank: mov @r0,#0ffh ;initialize the
pointed location by null address
      dec r0
      cjne r0,#2fh,blank
      mov r1,#41h    ;load address-
pointer with initial address

      mov 50h,#0dfh ;address for 16th
Display (rightmost)
      mov 4fh,#0bfh ;address for 15th
Display
      mov 4eh,#9fh  ;address for 14th
Display
      mov 4dh,#7fh  ;address for 13th
Display
      mov 4ch,#5fh  ;address for 12th
Display
      mov 4bh,#3fh  ;address for 11th
Display
      mov 4ah,#1fh  ;address for 10th
Display
      mov 49h,#0fbh ;address for 9th
Display
      mov 48h,#0f7h ;address for 8th
Display
      mov 47h,#0f3h ;address for 7th
Display
      mov 46h,#0efh ;address for 6th
Display
      mov 45h,#0ebh ;address for 5th
Display
      mov 44h,#0e7h ;address for 4th
Display
      mov 43h,#0e3h ;address for 3rd
Display
      mov 42h,#0fdh ;address for 2nd
Display
      mov 41h,#0feh ;address for 1st
Display (leftmost)

  chk:  mov a,input    ;load accumulator
with value at P3
        orl a,#0fh     ;mask lower nible
to get selection bus value
        cjne a,#0ffh,chk0
        mov dptr,#default ;load dptr
with starting address of defalt message
        sjmp read      ; now start
reading

  chk0: cjne a,#0fh,chk1
        mov dptr,#msg0 ;load dptr with
starting address of msg0
        sjmp read      ; now start
reading

  chk1: cjne a,#1fh,chk2
        mov dptr,#msg1
        sjmp read

  chk2: cjne a,#2fh,chk3
        mov dptr,#msg2
        sjmp read

  chk3: cjne a,#3fh,chk4
        mov dptr,#msg3
        sjmp read

  chk4: cjne a,#4fh,chk5
        mov dptr,#msg4
        sjmp read

  chk5: cjne a,#5fh,chk6
        mov dptr,#msg5
        sjmp read

  chk6: cjne a,#6fh,chk7
        mov dptr,#msg6
        sjmp read

  chk7: cjne a,#7fh,chk8
        mov dptr,#msg7
        sjmp read

  chk8: cjne a,#8fh,chk9
        mov dptr,#msg8
        sjmp read

  chk9: cjne a,#9fh,chk10
        mov dptr,#msg9
        sjmp read

 chk10: cjne a,#0afh,chk11
        mov dptr,#msg10
        sjmp read

 chk11: cjne a,#0bfh,chk12
        mov dptr,#msg11
        sjmp read

 chk12: cjne a,#0cfh,chk13
        mov dptr,#msg12
        sjmp read

 chk13: cjne a,#0dfh,chk14
        mov dptr,#msg13
        sjmp read

 chk14: mov dptr,#msg14
        sjmp read

 read: mov r3,dph
       mov r2,dpl
       setb tr0
  rd1: mov r0,01h
  rd2: clr a
       movc a,@a+dptr
       jz down
       mov DBH,a
       clr a
       inc dptr
       movc a,@a+dptr
       mov DBL,a
       mov ADB,@r0
       acall timer
       dec r0
       inc dptr
       sjmp rd2
 down: mov dph,r3 ;reload dph
       mov dpl,r2 ;reload dpl
       sjmp rd1

 timer: mov tmod,#10h ;set mode 1 for
timer1
       mov th1,#0fch ;FC66H will gen-
erate a delay of 1ms with 11.0592MHz
Xtal
       mov tl1,#66h
       setb tr1
       jnb tf1,$ ;wait until timer1
overflows
       clr tr1
       clr tf1
       mov DBH,#0ffh
       mov DBL,#0ffh
       ret

 ;** look-up table starts from here:
  msg0:   db Hh,Hl,Ah,Al,Ph,Pl,Ph,Pl,Yh
,Yl,bsh,bsl,Bh,Bl,Ih,Il,Rh,Rlw,Th,Tl,Hh
,Hl,bsh,bsl,Dh,Dl,Ah,Al,Yh,Yl,0
  msg1:   db Hh,Hl,Ah,Al,Ph,Pl,Ph,Pl,Y
h,Yl,bsh,bsl,Nh,Nl,Eh,El,Wh,Wl,bsh,bsl,
Yh,Yl,Eh,El,Ah,Al,Rh,Rlw,0
  msg2:   db strh,strl,bsh,bsl,Hh,Hl,
Ah,Al,Ph,Pl,Ph,Pl,Yh,Yl,bsh,bsl,Dh,Dl,
Ih,Il,Wh,Wl,Ah,Al,Lh,Ll,Ih,Il,bsh,bsl,
strh,strl,0
  msg3:   db Mh,Ml,Eh,El,Rh,Rlw,Rh,Rlw,
Yh,Yl,bsh,bsl,Ch,Cl,Hh,Hl,Rh,Rlw,Ih,Il,
Sh,Sl,Th,Tl,Mh,Ml,Ah,Al,Sh,Sl,0
  msg4:   db strh,strl,bsh,bsl,Hh,Hl,Ah
,Al,Ph,Pl,Ph,Pl,Yh,Yl,bsh,bsl,Nh,Nl,Oh,O
l,Lh,Ll,Ih,Il,bsh,bsl,strh,strl,0
  msg5:   db strh,strl,bsh,bsl,Eh,El,
Ih,Il,Dh,Dl,bsh,bsl,Mh,Ml,Uh,Ul,Bh,Bl,
Ah,Al,Rh,Rlw,Ah,Al,Kh,Kl,bsh,bsl,strh,
strl,0
  msg6:   db Hh,Hl,Ah,Al,Ph,Pl,Ph,Pl,Y
h,Yl,bsh,bsl,Dh,Dl,Ah,Al,Sh,Sl,Hh,Hl,Eh
,El,Hh,Hl,Rh,Rlw,Ah,Al,0
  msg7:   db Hh,Hl,Ah,Al,Ph,Pl,Ph,Pl,Y
h,Yl,bsh,bsl,Wh,Wl,Eh,El,Dh,Dl,Dh,Dl,Ih
,Il,Mh,Ml,Gh,Gl,0
  msg8:   db Hh,Hl,Ah,Al,Ph,Pl,Ph,Pl,Yh
,Yl,bsh,bsl,Jh,Jl,Ah,Al,Nh,Nl,Mh,Ml,Ah,A
l,Sh,Sl,Hh,Hl,Th,Tl,Mh,Ml,Ih,Il,0
  msg9:   db strh,strl,bsh,bsl,Hh,Hl,
Ah,Al,Ph,Pl,Ph,Pl,Yh,Yl,bsh,bsl,Rh,Rlw
,Ah,Al,Kh,Kl,Hh,Hl,Ih,Il,bsh,bsl,strh,
strl,0
  msg10:  db strh,strl,bsh,bsl,Hh,Hl,
Ah,Al,Ph,Pl,Ph,Pl,Yh,Yl,bsh,bsl,Ph,Pl,
Oh,Ol,Nh,Nl,Gh,Gl,Ah,Al,Lh,Ll,bsh,bsl,
strh,strl,0
  msg11:  db Hh,Hl,Ah,Al,Ph,Pl,Ph,Pl,Yh
,Yl,bsh,bsl,Mh,Ml,Oh,Ol,Th,Tl,Hh,Hl,Eh,E
l,Rh,Rlw,Sh,Sl,Dh,Dl,Ah,Al,Yh,Yl,0
  msg12:  db strh,strl,bsh,bsl,Hh,Hl,
Ah,Al,Ph,Pl,Ph,Pl,Yh,Yl,bsh,bsl,Rh,Rlw
,Ah,Al,Mh,Ml,Jh,Jl,Ah,Al,Nh,Nl,bsh,bsl
,strh,strl,0
  msg13:  db strh,strl,bsh,bsl,Hh,Hl,
Ah,Al,Ph,Pl,Ph,Pl,Yh,Yl,bsh,bsl,Lh,Ll,
Oh,Ol,Hh,Hl,Rh,Rlw,Ih,Il,bsh,bsl,strh,
strl,0
  msg14:  db strh,strl,bsh,bsl,Hh,Hl,
Ah,Al,Ph,Pl,Ph,Pl,Yh,Yl,bsh,bsl,Eh,El,
Ah,Al,Sh,Sl,Th,Tl,Eh,El,Rh,Rlw,bsh,bsl
,strh,strl,0
  default: db Wh,Wl,Eh,El,Lh,Ll,Ch,Cl,O
h,Ol,Mh,Ml,Eh,El,bsh,bsl,Th,Tl,Oh,Ol,bs
h,bsl,Ah,Al,Lh,Ll,Lh,Ll,0
       end
```